# An Optimization Problem with a Volume Constraint

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## 1 Introduction

These notes were created, in part, for a talk at the Grad ($\nabla$) Analysis Seminar at Rutgers on March 30th, 2026. I also decided to produce them as part of my reading this semester in preparation to do research on water waves and other problems in shape optimization. The notes are based chiefly on the 1986 paper *An Optimization Problem with a Volume Constraint* by Alt, Caffarelli, and Aguilera, with frequent references to the landmark 1981 paper *Existence and regularity for a minimum problem with free boundary* by Alt and Caffarelli. Please let me know if there are any errors.

## 2 The Constrained Problem

The classical conductor problem can be stated as follows: given a perfect conductor $\Gamma_0 = \partial\Omega$, where $\Omega$ is bounded or unbounded, we want to find a second one $\Gamma_1 = \Omega \cap \partial D$, where $D$ is a subdomain of $\Omega$ containing $\Gamma_0$ as its boundary so that their mutual energy is minimized along all $D$ with prescribed volume $\omega_0$. This particular problem with a volume constraint can be stated in the following way:

**Problem 2.1.** *Let $\Omega \subset \mathbb{R}^n$ be a domain with bounded Lipschitz boundary, $\Omega$ itself can be bounded or unbounded, and $u_0 \in H^1(\Omega)$ is nonnegative and given. We require that $u_0 > 0$ in a neighborhood of at least one $C^2$ regular point of $\partial\Omega$, and*

$$0 < \omega_0 < \lambda^n(\Omega)$$

*where $\lambda^n(\cdot)$ denotes n-dimensional Lebesgue measure. We look for an admissible minimizer $u \in H^1$ such that*

$$\begin{cases} u = u_0 \quad \text{on } \partial\Omega \\ \lambda^n(\{u > 0\}) = \omega_0 \end{cases}$$

*and*

$$E(u) := \int_\Omega |\nabla u|^2 \leq \int_\Omega |\nabla v|^2 =: E(v)$$

*where*

$$v \in \{w \in L^1_{loc}(\Omega); \nabla w \in L^2(\Omega), w = u_0 \text{ on } \partial\Omega, w \geq 0, \text{ and } \lambda^n(\{w > 0\}) = \omega_0\}$$

.

As we will see, proving the existence of a weak solution is not the difficult part. In fact, establishing the regularity of $u$ and proving that the free boundary $\partial\{u > 0\}$ is well behaved, in the same sense as the original Alt-Caffarelli problem, is the object of the paper.
That being said, though it is subtle, we will see the failure of the direct method when we try to pass to a minimizing subsequence immediately. This is the "proof" using direct method that we would expect.

**Proposition 2.1.** *There exists a weak solution $u \in H^1(\Omega)$ to Problem 1.1.*

*"Proof".* First, $E(v) = \int_\Omega |\nabla u|^2 \geq 0 \implies \inf_{K_0} E(v) = M$ exists. Now pick a sequence $\{u_j\} \in K_0$ such that $\lim_{j\to\infty} \int_\Omega |\nabla u_j|^2 = M$, and since since this sequence is in the class $K_0$, $|\nabla u_j| \leq C$ and $u_j = u_0$ on $\partial\Omega$ is the fixed boundary data. By Poincare,

$$\|u_j - \frac{1}{|\Omega|}\int_\Omega u_j\|_{L^2(\Omega)} \leq C\|\nabla u_j\|_{L^2(\Omega)} \leq C$$

and so $\{u_j\}$ are bounded in $H^1(\Omega)$. Thus, by Banach-Alaoglu, $\exists u_{j_k} \to u$ in $H^1(\Omega)$ weakly, and thus strongly in $L^2(\Omega)$ by Rellich-Kondrachov compactness. Now as $E$ is a convex functional, $E$ is lower semicontinuous, so

$$\int_\Omega |\nabla u|^2 \leq \liminf_{j\to\infty} \int_\Omega |\nabla u_j|^2 = M$$

so $u$ is a candidate for the minimum.
Now we must show that $u$ satisfies the volume constraint. We have that the n-dimensional Lebesgue measure is lower semicontinuous, so

$$\lambda^n(\{u > 0\}) \leq \liminf_{j\to\infty} \lambda^n(\{u_j > 0\}) = \omega_0$$

Suppose that $\lambda^n(\{u > 0\}) < \omega_0$. We want to show a contradiction by producing some $\tilde{u} \in H^1(\Omega)$ with less energy. Pick $x_0 \in \Gamma_1 = \partial\{u > 0\} \cap \Omega$, and define $D_\delta$ by expanding $\{u > 0\}$ around $B_\delta(x_0)$ such that $\lambda^n(D_\delta) = \omega_0$. We define $\tilde{u}$ such that $\Delta\tilde{u} = 0$ on $D_\delta$, $\tilde{u}|_{\partial\Omega} = u_0|_{\partial\Omega}$. and $\tilde{u} = u$ away from the expansion area, that is, on $D_\delta \setminus \{u > 0\}$. The loss in energy is given by Hadamard's variational formula:

$$\delta E = -\int_{D_\delta \setminus \{u>0\}} |\nabla\tilde{u}|^2 dx$$

If $|\nabla u| > 0$ on $\partial\{u > 0\}$, then expanding $\{u > 0\}$ even slightly reduces $E$, and thus

$$\int_\Omega |\nabla\tilde{u}|^2 < \int_\Omega |\nabla u|^2$$

which contradicts the fact that $u$ was a minimizer. Therefore, $\lambda^n(\{u > 0\}) = \omega_0$. □

What is the problem with this attempt? The class

$$K_0 = \{w \in L^1_{loc}(\Omega); \nabla w \in L^2(\Omega), w = u_0 \text{ on } \partial\Omega, w \geq 0, \text{ and } \lambda^n(\{w > 0\}) = w_0\}$$

is not even weakly closed! To see why, we can consider a classical counterexample. Let $\Omega = [0,1] \times [0,1] \subset \mathbb{R}^2$, and let $\{u_j\} \in H^1(\Omega)$. Let each $u_j$ be supported on $j$ separate disjoint islands $\{I_n\}_{1 \leq n \leq j}$ of the same area, such that

$$\lambda^2(\cup_{k=1}^{j} I_k) = \lambda^2(\text{supp}(u_j)) = \frac{1}{2}$$

for every $j$. So as $j$ becomes large, the radius $r_j$ of each island goes to 0, and the maximum value of the function the set is bounded and goes to 0 as well as $j \to \infty$. What we obtain is

$$u_j \to u = 0 \quad \text{a.e.}$$

and thus $u_j \rightharpoonup 0$ weakly in $H^1(\Omega)$, and so $\lambda^2(\text{supp}(u)) = 0$, yet $\lambda^2(\text{supp}(u_j)) = \frac{1}{2}$ for all $j \in \mathbb{N}$, $K_0$ is not weakly closed. Thus, we need some different machinery from the calculus of variations to tackle this problem.

# 3 The Unconstrained Problem

We will call Problem 1.1 the *constrained* problem, and we would like to formulate the same minimization problem in terms of minimizing an *unconstrained* energy functional. This is were we use the free boundary condition. By Hadamard, if $\delta\rho$ is the displacement along the boundary, then

$$\delta E = -\int_{\Gamma_1} (\partial_\nu u)^2 \delta\rho d\sigma$$

where $\Gamma_1 = \partial\{u > 0\} \cap \Omega$. Further,

$$\delta\text{Vol} = \int_{\Gamma_1} \delta\rho d\sigma$$

Now, we would like to employ a Lagrange multiplier method to balance the change of energy with the change in volume. For some $\lambda > 0$, we have

$$\begin{aligned} \delta E + \lambda^2 \delta\text{Vol} &= 0 \\ \implies \int_{\Gamma_1} (-(\partial_\nu u)^2 + \lambda^2)\delta\rho d\sigma &= 0 \\ \implies -(\partial_\nu u)^2 + \lambda^2 &= 0 \\ \implies -\partial_\nu u &= \lambda \end{aligned}$$

on $\partial\{u = 0\} \cap \Omega$. Such a solution is found by minimizing the modified energy

$$\tilde{E}_\lambda(u) = \int_\Omega |\nabla u|^2 + \lambda^2 \chi_{\{u>0\}}\, dx$$

on the set $K := \{v \in L^1_{loc}(\Omega) : \nabla v \in L^2(\Omega), v = u_0 \text{ on } \partial\Omega, v \geq 0\}$. Note that by Alt-Caffarelli, minimzers $\tilde{u}_\lambda$ of $\tilde{E}_\lambda$ are Lipschitz continuous, and $\partial\{\tilde{u}_\lambda > 0\} \cap \Omega$ is an analytic hypersurface on which $-\partial_\nu \tilde{u}_\lambda = \lambda$ is satisfied.
Here is the dilemma. For a choice of $\lambda$, if $\lambda^n(\{\tilde{u}_\lambda > 0\}) = \omega_0$, then $\tilde{u}_\lambda$ becomes a minimizer of $E$. However, given $\omega_0$, it is not certain that the corresponding $\lambda$ even exists.

### 3.1 Penalization Functional

To obviate this problem, we introduce the following functional:

**Definition 3.1** (Penalization Functional)**.** *Let*

$$E_\epsilon(v) = \int_\Omega |\nabla v|^2 + f_\epsilon(\lambda^n(\{v > 0\})$$

*with*

$$f_\epsilon(s) = \begin{cases} \frac{1}{\epsilon}(s - \omega_0), & s \geq w_0 \\ \epsilon(s - \omega_0), & s \leq w_0 \end{cases}$$

The idea behind this functional is the following. Trying to make volume preserving transformations of the positivity set $D := \{v > 0\}$ is far too analytically taxing. This would involve pushing one part of $D$ out while having to pull a different part in to maintain the same volume. Instead, this functional perfectly accounts for nonvolume preserving transformations, that is, only pushing one part of the domain $D$ out and increasing its size.
If $s \geq \omega_0$, then the quantity $f_\epsilon$ becomes massive, so $E_\epsilon$ heavily penalizes the size of the positivity set being any greater than the volume constraint. If instead $s \leq \omega_0$, then the quanity $f_\epsilon$ becomes negative, which seems like it would only decrease $f_\epsilon$. However, we are saved by the fact that the Dirichlet energy increases at a faster rate as we shrink the positivity set.
From now on, $u_\epsilon$ will refer to a minimizer of the penalization functional $E_\epsilon$.

## 4 The Alt-Caffarelli Properties for $u_\epsilon$

One object of this paper is to show that minimizers $u_\epsilon$ of $E_\epsilon$ are weak solutions of the free boundary problem

$$-\partial_n u_\epsilon = \lambda_\epsilon$$

for some $\lambda_\epsilon$ in the sense of Alt-Caffarelli (what does this mean precisely?), so that the theory from the famous 1981 paper can be applied. More specifically, Alt-Caffarelli proved that minimizers to unconstrained functionals of the type

$$E(u) = \int_\Omega |\nabla u|^2 + \Lambda \chi_{u>0}$$

are weak solutions of the corresponding Bernoulli problem

$$\begin{cases} \Delta u = 0 \quad \text{in } \{u > 0\} \\ |\nabla u| = \sqrt{\Lambda} \quad \text{on } \partial\{u > 0\} \end{cases}$$

and satisfy the following three conditions:

(1) Lipschitz continuity

(2) Nondegeneracy

(3) Positive Density

We will expand on what these three conditions mean precisely in the context of this problem in a moment. First, let us note something remarkable about the sequence $\{u_\epsilon\}$ as it pertains to the volume constraint.

**Proposition 4.1.** *For $\epsilon$ small enough (not necessarily passing to a limit), if $\{u_\epsilon\}$ is a minimizer of $E_\epsilon$, then*

$$\lambda^n(\{u_\epsilon > 0\}) = \omega_0$$

*Proof.* First, we will have to black box a result which will be proved later, that is, there exists constants $c, C$ independent of $\epsilon$ such that

$$0 < c \leq \lambda_\epsilon \leq C < \infty$$

While this is a reasonable claim, it requires a bit of work to prove.
Intuitively, since $f'_\epsilon(s)$ jumps from a small to large value at $s = w_0$, we would expect that $\lambda^n(\{u_\epsilon > 0\}) = \omega_0$. To justify this, Suppose $\lambda^n(\{u_\epsilon > 0\}) > \omega_0$, and induce an inward perturbation with displacement $\rho$ and volume change $\delta V = \int_{\partial\{u_\epsilon>0\}\cap\Omega} \delta\rho d\sigma < 0$. We obtain that the change in $f_\epsilon$ is given by

$$\delta f_\epsilon = \delta\left(\frac{1}{\epsilon}((\lambda^n(\{u_\epsilon > 0\}) - \omega_0)\right) = \frac{1}{\epsilon}\delta\left(\lambda^n(\{u_\epsilon > 0\})\right) = -\frac{1}{\epsilon}\delta V$$

Moreover, by Hadamard, the Dirichlet energy will change like

$$\delta E = \int_{\partial\{u_\epsilon>0\}\cap\Omega} (\partial_\nu u_\epsilon)^2 \delta\rho d\sigma = \lambda_\epsilon^2 \int_{\partial\{u_\epsilon>0\}\cap\Omega} \delta\rho d\sigma = \lambda_\epsilon^2 \delta V \leq C^2 \delta V$$

Since $u_\epsilon$ is a minimizer of $E_\epsilon$, we require that the change in energy exceeds the change in the volume penalty. However, for $\epsilon$ small enough such $\frac{1}{\epsilon} > C^2$, we have that

$$\delta E_\epsilon = \delta E + \delta f_\epsilon \leq C^2\delta V - \frac{1}{\epsilon}\delta V = \left(C^2 - \frac{1}{\epsilon}\right)\delta V < 0$$

This is a contraction to $u_\epsilon$ being a minimzer.
Assume instead that $\lambda^n(\{u_\epsilon > 0\}) < \omega_0$. In this case, we perform an outward perturbation instead, pushing $\partial\{u_\epsilon > 0\}$ out with displacement $\rho$, such that

$$\delta V = \int_{\partial\{u_\epsilon>0\}\cap\Omega} \delta\rho d\sigma > 0$$

We now obtain that $\delta f_\epsilon = \epsilon\delta V$, and the change (loss) of energy is now

$$\delta E = -\lambda_\epsilon^2 \delta V \leq c^2 \delta V$$

If $\epsilon$ is small enough such that $\epsilon < c^2$, then we have that

$$\delta E_\epsilon = \delta E + \delta f_\epsilon \leq -c^2\delta V + \epsilon\delta V = (\epsilon - c^2)\delta V < 0$$

which also contradicts $u_\epsilon$ being a minimizer. □

**Theorem 4.1.** *There exists a minimizer $u_\epsilon$ with the following properties:*

*(1) :* $u_\epsilon \in C^{0,1}_{loc}(\Omega)$

*(2) :* $c dist(x, \partial\{u_\epsilon > 0\}) \leq u_\epsilon(x) \leq C dist(x, \partial\{u_\epsilon > 0\})$ *for* $x \in D$

*(3) :* $c \leq \frac{\lambda^n(B_r(x) \cap \{u_\epsilon > 0\})}{\lambda^n(B_r(x))} \leq 1 - c$

*where $D \subset\subset \Omega$ is any subdomain, and $c, C$ depend on $D$ and $\epsilon$.*

*Proof.* First, let

$$K := \{v \in H^1(\Omega) : v \geq 0 \text{ in } \Omega, v = u_0 \text{ on } \partial\Omega\}$$

for some prescribed function $u \in H^1(\Omega)$. If we choose $u$ with $\lambda^n(\{u > 0\}) \leq \omega_0$, we obtain that $E_\epsilon(u) \leq C$ uniformly in $\epsilon$ since

$$E_\epsilon(u) = \int_\Omega |\nabla u|^2 dx + f_\epsilon(s) \leq c_0 + \epsilon(s - \omega_0) \leq c_0$$

and $E_\epsilon(v) \geq -\omega_0$ for $v \in K$ since for $s \leq \omega_0$, we have

$$E_\epsilon(v) \geq \epsilon(s - \omega_0) \geq -\epsilon\omega_0 \geq -\omega_0$$

for any $\epsilon \leq 1$. Thus, a minimizing sequence $\{u_k\}$ exists, and

$$\int_\Omega |\nabla u_k|^2 \leq C, \quad \lambda^n(\{u_k > 0\}) \leq C$$

Thus a limit function $u_\epsilon \in K$ exists, and

$$u_k \rightharpoonup u \in H^1(\Omega)$$

and thus

$$u_k \to u_\epsilon \in L^2(\Omega)$$

strongly by Rellich-Kondrachov compactness. Moreover,

$$\lambda^n(\{u_\epsilon > 0\}) \leq \liminf_{k\to\infty} \lambda^n(\{u_k > 0\})$$

since, by the fact that $\chi$ is lower semicontinuous (Fatou's lemma),

$$\int_\Omega \chi_{\{u_\epsilon > 0\}} dx \leq \int_\Omega \liminf_{k\to\infty} \chi_{\{u_k > 0\}} dx \leq \liminf_{k\to\infty} \int_\Omega \chi_{\{u_k > 0\}} dx$$

For obtaining (2) and (3), we need to use the fact

$$\epsilon(s_2 - s_1) \leq f_\epsilon(s_2 - s_1) \leq \frac{1}{\epsilon}(s_2 - s_1)$$

The idea is this, for $x_0 \in \partial\{u_\epsilon > 0\}$ we define $h$ to be the harmonic replacement on $B_r(x_0)$ with the same boundary data as $u_\epsilon$. We have that

$$\int_{B_r(x_0)} |\nabla u_\epsilon|^2 + f_\epsilon(\lambda^n\{u_\epsilon > 0\}) \leq \int_{B_r(x_0)} |\nabla h|^2 + f_\epsilon(\lambda^n\{h > 0\})$$

$$\implies \int_{B_r(x_0)} |\nabla u_\epsilon|^2 - |\nabla h|^2 \leq \frac{1}{\epsilon}\lambda^n(B_r \cap \{u = 0\})$$

which is the key estimate as in Alt-Caffarelli □

# 5 Behavior of $\{u_\epsilon > 0\}$ for small $\epsilon$

**Proposition 5.1.** *There exists constants $c, C$ independent of $\epsilon$ such that*

$$0 < c \leq \lambda^n(\{u_\epsilon > 0\}) \leq \omega_0 + C\epsilon$$

*Proof.* From Theorem 3.1:

$$E_\epsilon(u_\epsilon) \leq E_\epsilon(u_0) \leq C$$

independent of $\epsilon$, so

$$f_\epsilon(\lambda^n(\{u_\epsilon > 0\}) \leq C \implies \frac{1}{\epsilon}(\lambda^n(\{u_\epsilon > 0\} - \omega_0) \leq C$$

$$\implies \lambda^n(\{u_\epsilon > 0\}) \leq C\epsilon + \omega_0$$

Which establishes the upper bound. Note that $\int_\Omega |\nabla u_\epsilon|^2 \leq C$, and $u_\epsilon \equiv u_0$ on $\partial\Omega$. To get the lower bound, we argue by integrating on lines with $D := \Omega \cap B_\delta(\partial\Omega)$, a thin tubular neighborhood around $\partial\Omega$. We parametrize $D$ using $(x', t)$ where $x' \in \partial D$, $t \in (0, \delta)$, so that $u_\epsilon(x', 0) = u_0(x', 0) = u_0|_{\partial\Omega}$. By the fundamental theorem of calculus, we have

$$u_0(x') = u_\epsilon(x', t) - \int_0^t \frac{\partial u_\epsilon}{\partial \nu}(x', s) ds$$

Integrating both sides with respect to $t$ from 0 to $\delta$ and taking absolute values yields

$$\delta |u_0(x')| \leq \int_0^\delta |u_\epsilon(x', t)| dt + \delta \int_0^\delta \left| \frac{\partial u_\epsilon}{\partial \nu}(x', s) \right| ds$$

$$\implies \delta \int_{\partial\Omega} u_0(x') \leq \int_D |u_\epsilon| dx + \delta \int_D |\nabla u_\epsilon| dx$$

Let $D^+ := D \cap \{u_\epsilon > 0\}$, and since $u_\epsilon = 0$ outside $D^+$, can replace the domain of integration for both terms on the right hand side by $D^+$. By Cauchy-Schwarz, we have that

$$\delta \int_{\partial\Omega} u_0(x') \leq (\lambda^n(D^+))^{1/2} \left( \left( \int_D |u_\epsilon|^2 dx \right)^{1/2} + \delta \left( \int_D |\nabla u_\epsilon|^2 dx \right)^{1/2} \right)$$

Squaring both sides and using the algebraic identity $(a+b)^2 \leq 2a^2 + 2b^2$, we have

$$\delta^2 \left( \int_{\partial\Omega} u_0 \right)^2 \leq 2\lambda^n(D \cap \{u_\epsilon > 0\}) \left( \int_D |u_\epsilon|^2 dx + \delta^2 \int_D |\nabla u_\epsilon|^2 dx \right)$$

$$\implies \left( \int_{\partial\Omega} u_0 \right)^2 \leq 2\lambda^n(D \cap \{u_\epsilon > 0\}) \left( \frac{1}{\delta^2} \int_D |u_\epsilon|^2 dx + \int_D |\nabla u_\epsilon|^2 dx \right)$$

$$\implies \left( \int_{\partial\Omega} u_0 \right)^2 \leq 2 \max\left(1, \frac{1}{\delta^2}\right) \lambda^n(D \cap \{u_\epsilon > 0\}) \left( \int_D |u_\epsilon|^2 dx + \int_D |\nabla u_\epsilon|^2 dx \right)$$

$$\implies \left( \int_{\partial\Omega} u_0 \right)^2 \leq C(\delta) \lambda^n(D \cap \{u_\epsilon > 0\}) \left( \int_D |u_\epsilon|^2 dx + \int_D |\nabla u_\epsilon|^2 dx \right)$$

$$\implies \frac{||u_0||^2_{L^2(\partial\Omega)}}{C(\delta)||u_\epsilon||^2_{H^1(D)}} \leq \lambda^n(D \cap \{u_\epsilon > 0\}) \leq \lambda^n(\{u_\epsilon > 0\})$$

which establishes the lower bound $\lambda^n(\{u_\epsilon > 0\}) \geq c$. □

**Lemma 5.1.** *$\lambda_\epsilon \leq C$, where $C$ is independent of $\epsilon$.*

*Proof.* We will begin by proving the case where $\Omega$ is bounded. Let $w$ be the harmonic function in $\Omega$ with $w|_{\partial\Omega} \equiv u_0|_{\partial\Omega}$. We now define the set

$$D := \Omega \setminus \bar{B}_\delta(\partial\Omega)$$

and $\delta$ is to be chosen such that

$$\omega_1 := \lambda^n(D) > \omega_0$$

and

$$\lambda^n(\Omega \setminus D) < c$$

with the same $c$ as in Proposition 4.1. With this, we have

$$\lambda^n(D \cap \{u_\epsilon > 0\}) \leq \omega_0 + C\epsilon \leq \frac{\omega_0 + \omega_1}{2} < \omega_1 = \lambda^n(D)$$

and furthermore,

$$\lambda^n(D \cap \{u_\epsilon > 0\}) \geq \lambda^n(\{u_\epsilon > 0\}) - \lambda^n(\Omega \setminus D) \geq c - \lambda^n(\Omega \setminus D) > 0$$

since $D$ was chosen such that $\lambda^n(\Omega \setminus D) < c$. Let us recall a theorem in section 2 regarding harmonic measure, that is for any $\zeta \in C_0^\infty(\Omega)$, we have

$$\int_\Omega \nabla u_\epsilon \nabla \zeta \, dx = -\int_{\Omega \cap \partial\{u_\epsilon > 0\}} q_{u_\epsilon} \zeta \, d\mathcal{H}^{n-1}$$

In this case, while $u_\epsilon - w$ is merely $H_0^1(\Omega)$ apriori, we can assume instead that $\zeta := u_\epsilon - w$ is $C_0^\infty(\Omega)$ and argue by density to attain the desired result. Thus,

$$C \geq \int_\Omega \nabla(u_\epsilon - w)\nabla u_\epsilon \, dx = \int_\Omega \nabla\zeta\nabla u_\epsilon \, dx = -\int_{\Omega \cap \partial\{u_\epsilon > 0\}} \zeta q_{u_\epsilon} \, d\mathcal{H}^{n-1}$$

Also, by theorem 3 of section 2, we have that $q_{u_\epsilon} = \lambda_\epsilon \, \mathcal{H}^{n-1}$ almost everywhere on $\partial_{red}\{u_\epsilon > 0\}$, where $\partial_{red}$ denotes the reduced boundary, and so

$$\begin{aligned} C \geq & -\int_{\Omega \cap \partial\{u_\epsilon > 0\}} \zeta\lambda_\epsilon \, d\mathcal{H}^{n-1} \\ = & \int_{\Omega \cap \partial\{u_\epsilon > 0\}} w\lambda_\epsilon d\mathcal{H}^{n-1} \\ \geq & \lambda_\epsilon \inf_D w\mathcal{H}^{n-1}(D \cap \partial\{D \cap \{u_\epsilon > 0\})\} \\ \geq & c\lambda_\epsilon \implies \lambda_\epsilon \leq \frac{C}{c} \end{aligned}$$

which is what we desired to show.
For the unbounded case, we choose a fixed ball $B_R(0)$ containing $\partial\Omega$ with

$$\lambda^n(\Omega \cap B_R(0)) > \omega_0$$

Now, for the proof when $\Omega$ is unbounded as an exterior domain, for simplicity, we will assume that $u_0 > 0$ on $\partial\Omega$, and $\partial\Omega$ is smooth, the main idea is the following. $\Delta u = 0$ when $u$ is positive, so by divergence theorem

$$0 = \int_\Omega \Delta u_\epsilon \, dx = \int_{\partial\Omega \cup \partial\{u_\epsilon > 0\}} \partial_\nu u_\epsilon \, dx = \int_{\partial\Omega} \partial_\nu u_\epsilon + \int_{\partial\{u_\epsilon>0\}} \partial_\nu u_\epsilon$$

$$\implies \int_{\partial\Omega} \partial_\nu u_\epsilon \, dx = -\int_{\partial\{u_\epsilon>0\}} \partial_\nu u_\epsilon \, dx = \int_{\partial\{u_\epsilon>0\}} \partial_{-\nu} u_\epsilon \, dx$$

Thus, since $u_\epsilon$ has bounded energy, we have

$$C \geq \int_\Omega |\nabla u_\epsilon|^2 = \int_{\partial\Omega} u_0 \partial_\nu u_\epsilon \geq \inf_{\partial\Omega} u_0 \int_{\partial\Omega} \partial_\nu u_\epsilon$$

$$= \inf_{\partial\Omega} u_0 \int_{\Omega\cap\partial\{u_\epsilon>0\}} \partial_{-\nu} u_\epsilon \, d\mathcal{H}^{n-1} = \lambda_\epsilon \inf_{\partial\Omega} u_0 \mathcal{H}^{n-1}(\Omega \cap \partial\{u_\epsilon > 0\})$$

$$\implies C \geq c\lambda_\epsilon \implies \lambda_\epsilon \leq \frac{C}{c}$$

independent of $\epsilon$. □

**Lemma 5.2.** *$\lambda_\epsilon \geq c > 0$, with $c$ independent of $\epsilon$.*

*Proof.* The geometric setup is this: let $y_0 \in \partial\Omega$, $B_{\rho_0}(y_0)$ a ball such that $B_{\rho_0}(y_0) \cap \partial\Omega$ is $C^2$ and $u_0 \geq c_0 > 0$ in $B_{\rho_0}(y_0)$. Assume that $u_\epsilon > 0$ in a neighborhood of $y_0$, and let $y_1$ be any free boundary point of $u_\epsilon$ in $\Omega \setminus \{B_\delta(\partial\Omega)\}$, where $\delta$ is independent of $\epsilon$. Then we choose a family of smooth domains $D_t \subset \Omega \cup B_{\rho_0}$ such that $D_0 \subset B_{\rho_0}(y_0) \setminus \Omega$, touching $\partial\Omega$ at $y_0$ and $y_1 \subset D_1$. Let $t$ be the first value for which $D_t$ touches a free boundary point $x_0 \in \partial D_t \cap \partial\{u_\epsilon > 0\}$.
Now let $w$ be the harmonic function in $D_t \setminus \bar{D}_0$ with $w|_{\partial D_0} \equiv c_0$ and $w|_{\partial D_t} \equiv 0$. Then we have that

$$\partial_{-\nu} \geq cc_0$$

by Hopf's Lemma. Moreover, since $u_\epsilon \geq w$ on $\partial(D_t \cap \Omega)$, we have by comparison principle that $u \geq w$ also in $D_t \cap \Omega$. Therefore, if $x_0$ is a regular free boundary point $\lambda_\epsilon = \partial_{-\nu} u_\epsilon(x_0) \geq \partial_{-\nu} w(x_0) \geq c > 0$, where $c$ is independent of $\epsilon$. However, there is more work to be done if we cannot be certain $x_0$ is a regular free boundary point. For small $r > 0$, we have

$$⨍_{\partial B_r(x_0)} u \geq ⨍_{\partial_r(x_0)} w \geq cr$$

Now let $v_0$ be the harmonic replacement of $u_\epsilon$ on $\partial B_r$:

$$\begin{cases} \Delta v_0 = 0 \text{ in } \Omega \\ v_0 = u_\epsilon \text{ on } \partial\Omega \end{cases}$$

From Alt-Caffarelli, we have that

$$\int_{B_r(x_0)} (|\nabla u|^2 - |\nabla v_0|^2) = \int_{B_r(x_0)} |\nabla(u - v_0)|^2 \geq c\lambda^n(B_r(x_0) \cap \{u = 0\})$$

where $c$ is independent of $\epsilon$. Now let $x_1$ be a free boundary point away from $x_0$. We can choose $x_1$ to be regular so tat $\partial\{u_\epsilon > 0\}$ is smooth in $B_{r_0}(x_1)$ for some small $r_0$. Near $x_1$ we make a smooth perturbation of $\{u_\epsilon > 0\}$ decreasing its volume by $\delta_r := \lambda^n(B_r(x_0) \cap \{u = 0\})$. Let $v_1$ be the harmonic function in the perturbed set vanishing on its boundary and equal to $u$ on $\partial B_{r_0}(x_1)$. Then we have that

$$\int_{B_{r_0}(x_1)} (|\nabla v_1|^2 - |\nabla u|^2) = \lambda_\epsilon^2 \delta_r + o(\delta_r)$$

Because the function

$$v := \begin{cases} v_0 & \text{in } B_r(x_0) \\ v_1 & \text{in } B_{r_0}(x_0) \\ u_\epsilon & \text{elsewhere} \end{cases}$$

satisfies $\lambda(\{v > 0\}) = \lambda(\{u_\epsilon > 0\})$, we obtain

$$\begin{aligned} 0 \leq E_\epsilon(v) - E_\epsilon(u) &= \int_{B_r(x_0)} |\nabla v_0|^2 - |\nabla u|^2 + \int_{B_r(x_0)} (|\nabla v_1|^2 - |\nabla u|^2) \\ &\leq -c\delta_r + \lambda_\epsilon^2 \delta_r + O(\delta_r) \end{aligned}$$

Therefore, $\lambda_\epsilon \geq c$. □